

MOHD. FARHAN ISRAK SOUMIK

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RESEARCH INTERESTS

My research examines the security, privacy, and forensic challenges of human interaction with large-scale, consumer-facing generative AI systems, with an emphasis on building trustworthy and human-centered AI technologies. I focus on understanding privacy risks arising from implicit behavioral signals and on defending against malicious or deceptive uses of generative AI that threaten user safety and public trust.

Technical Focus Areas: Federated learning; large language models (LLMs); vision-language models (VLMs); multimodal AI; privacy-preserving machine learning; generative AI forensics.

CAREER GOALS

• **Short-term Goal:** To complete my Ph.D. with a strong record of high-impact research in secure, privacy-preserving, and trustworthy AI, developing an independent research agenda at the intersection of generative AI, multimodal learning, and human-centered security.

• **Long-Term Goal:** To pursue a faculty position in computer science, where I can lead an interdisciplinary research program, mentor graduate and undergraduate students, and contribute through teaching, scholarship, and service to advancing responsible and trustworthy AI systems.

EDUCATION

Southern Illinois University Carbondale, IL, USA

2024 -

Ph.D. in Computer Science CGPA 4.00 (on the scale of 4.00)

Supervised by: Dr. Abdur Rahman Bin Shahid, Assitant Professor & Director of SHIELD Lab, Dept. of Computer Science and Engineering, SIUC.

Rajshahi University of Engineering & Technology(RUET), Bangladesh

2016 - 2020

B.Sc in Computer Science & Engineering CGPA 3.42 (on the scale of 4.00)

Thesis: An analysis on the impact of deep learning based models for classification and segmentation of tumors from medical images.

Supervised by: Dr. Md. Ali Hossain, Professor, Dept. of Computer Science and Engineering, RUET.

SKILLS

Core Technical Areas	Machine Learning and Multimodal Learning; Computer Vision; Signal Processing; Security and Privacy of AI Systems; Data Science
AI Skills	Generative AI; Large Language Models (LLMs); Vision-Language Models (VLMs); Agentic AI; Privacy-Preserving AI; AI Forensics
Programming Skills	Python, Matlab, R, Java, C, C++, Assembly
Frameworks & Libraries	Tensorflow,Keras, Pytorch, Sckit-learn, Pandas, OpenCV

EXPERIENCE

Graduate Research Assistant

June 2025-Current

Secure and Trustworthy Intelligent Systems Lab([SHIELD Lab](#)), SIUC

Carbondale, Illinois, USA

- Conducting doctoral research focusing on large language models, vision-language models, and usable privacy and security.

- **Key Achievements:**

- Paper having the title *"No Innocence in Styling-Discovery of Privacy Protection Capabilities and Security Risks in Consumer Generative AI Writing Assistants"* accepted for Presentation at **ACL Industry Track-2026** at San Diego, California.

- Published a research paper, "*Privacy-Preserving Multimodal Stress Detection from Wearables with Attention Fusion and Federated Learning*" at **25th International Conference on Machine Learning and Applications(ICMLA-2025)** at Boca Raton, Florida.
- Published a research paper on privacy preservation in human interaction with generative AI at **PPAI-25: The 6th AAAI Workshop on Privacy-Preserving Artificial Intelligence**, co-located with the **AAAI Conference on Artificial Intelligence 2025**, one of the top venues in AI research across journals and conferences and sponsored by organizations including Google, Apple, and Deloitte.
- Awarded a competitive travel grant to present my research at the PPAI 2025 Workshop on Privacy-Preserving Artificial Intelligence, co-located with AAAI 2025.
- Developing the first large-scale dataset of generative AI-based image edits from major consumer platforms such as Apple Intelligence and Samsung AI, aimed at early detection of manipulated content and the development of user-facing tools to help diverse populations understand and identify AI-generated image edits.

Graduate Teaching Assistant

School of Computing, Southern Illinois University Carbondale(SIUC)

August 2024-June 2025

Carbondale, Illinois, USA

- Held 20 hours/week of office hours supporting undergraduate students, and assisted faculty with preparing quizzes and assignments and grading exams and coursework.

Research Assistant

Center for Computational and Data Sciences(CCDS), Independent University ([Link](#))

Jan 2022 - June 2023

Dhaka, Bangladesh

- Worked on developing data level fusion model for lulc classification of satellite images as well as developed state of the art segmentation models for LULC using Synthetic Aperture Radar(SAR) images as part of a collaboration between CCDS and Data & Design Lab([DND Lab](#)) of University of Dhaka

Teaching Assistant

Independent University([IUB](#))

Jan 2022 - June 2023

Dhaka, Bangladesh

- Duties involved grading and holding class. Assigned courses were CSC-101 Introduction to Programming, CSC-203 Data Structures, CIS-101 Fundamentals of Computer Systems.

GRE Instructor

Graduate Resources Enhancing Center([GREC](#))

June 2023 - Current

Dhaka, Bangladesh

- Conduct part-time instruction for Graduate Record Examination (GRE) preparation, teaching Quantitative, Verbal, and Analytical Writing sections.
- Designed and administered daily and weekly quizzes, and graded practice tests and full-length mock examinations.

Data Science and Business Analytics Intern

The Sparks Foundation

July 2021 - August 2021

Singapore City, Singapore

- Implemented different types of machine learning algorithms ranging from Random Forest to SVM together with sequence models like RNN and LSTM for solving different types of business related problems.

PUBLICATIONS

M. F. I. Soumik, W. K. M. Mithsara, A. R. Shahid, and A. Imteaj, "**Audio Editing Features as User-Centric Privacy Defenses Against Large Language Model (LLM)-Based Emotion Inference Attacks**," *Proceedings of the 6th AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI-25)* [[arXiv Link](#)]

M. F. I. Soumik, E. Maghsoudlou Nima, H. Zangoti, A. Imteaj, and A. R. Shahid, "**Privacy-Preserving Multimodal Stress Detection from Wearables with Attention Fusion and Federated Learning**," 2025. *Proceedings of 24th International Conference on Machine Learning Applications(ICMLA)*

M. F. I. Soumik, S. M. Hasan, and A. R. Shahid, “Evaluating Apple Intelligence’s Writing Tools for Privacy Against Large Language Model-Based Inference Attacks: Insights from Early Datasets,” *arXiv preprint arXiv:2506.03870*, 2025. [\[Link\]](#)

M. F. I. Soumik and M. A. Hossain, “Brain Tumor Classification with Inception Network Based Deep Learning Model Using Transfer Learning,” *Proceedings of the 2020 IEEE Region 10 Symposium (TENSYP)* [\[Link\]](#)

CERTIFICATIONS

- *Machine Learning by Stanford University.*[Link](#)
- *Deep Learning Specialization by DeepLearning.AI through Coursera.* [Link](#)
- *Generative AI With Large Language Models by Deep Learning.AI*[Link](#)
- *DeepLearning.AI TensorFlow Developer Professional Certificate.* [Link](#)
- *Mathematics for Machine Learning Specialization by Imperial College of London.* [Link](#)

PROBLEM SOLVING

Solved more than **500** problems in different online Judges including Hackerrank, Leetcode, UVA. Achieved Gold badge for problem solving in Hackerrank [Link](#).

AWARDS & ACHIEVEMENTS

- Travel Grant for Poster Presentation at 6th AAAI workshop on Privacy Preserving AI (PPAI-2025)
- Government Scholarship and full tuition waiver for securing 85th place(Top 5%) in the merit list of Higher Secondary School Certificate Examination,2015

TEST SCORES

- Graduate Record Examination Score: **317(Quant-164,Verbal-153,AWA-4.00)**
- IELTS Score: **8.00/9.00(Listening-8.5, Reading-8.5, Writing-7.0,Speaking-7.00)**

REFERENCES

- Dr. Abdur Rahman Bin Shahid
Assistatn Professor,
Department of Computer Science, Southern Illinois University Carbondale(SIUC). Mail: shahid@cs.siu.edu
- Dr. Ahmed Imteaj
Assistant Professor
Department of Electrical Engineering and Computer Science, Florida Atlantic University. Mail: aimteaj@fau.edu